

Picture Graphs and Bar Graphs

Answer Key

Grade 2–3

Quick Answers

1. 20	3. 4	5. 41	7. 10
2. 12	4. 12	6. 7	8. —

Curriculum

Level: Grade 2–3

2.MD / 3.MD – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

4. If they got 6: counted grid lines instead of using the scale, so each line counted as 1 instead of 2
6. If they got 8: read the top of the bar at the nearest labelled line instead of halfway between two lines
7. If they got 12: counted 3 whole pictures and ignored that the last one is only half a picture
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What this sheet is teaching, and what to listen for. A younger child reads a graph by counting: one picture, one thing. A *scaled* graph breaks that habit on purpose — one picture is worth four books, one grid line is worth two bikes — and the child has to look at the key or the scale *before* counting anything. Almost every wrong answer on this sheet comes from skipping that look. When you hear a number that is much too small, do not correct it: point at the key and ask “what is one picture worth?”

1. How many books did Ana read?

Ana’s row has 5 book pictures. The key says one picture stands for 4 books, so the count goes in fours: 4, 8, 12, 16, 20.

Pictures: 5. Books: $5 \times 4 =$ 20.

If they say 5: they have counted pictures instead of books. Cover the row with your hand except the first picture and ask “how many books is just that one?” Then let them count again.

2. How many books did Ben read?

Ben’s row has 3 book pictures. Skip-counting by fours: 4, 8, 12.

Books Ben read: $3 \times 4 =$ 12.

The skip-count is the answer they should write down, not just the total. A child who can say “four, eight, twelve” while touching each picture has the idea; a child who writes 12 with no working may have copied it from a neighbour’s sheet or guessed.

3. How many more books did Cara read than Ana?

Two steps, and the first one is easy to skip.

Cara has 6 pictures: $6 \times 4 = \boxed{24}$ books. Ana has 5 pictures: $5 \times 4 = \boxed{20}$ books.

Now compare the *books*, not the pictures: $24 - 20 = \boxed{4}$.

Watch for the answer 1. A child who compares the pictures ($6 - 5 = 1$) has compared the wrong quantity. The question asks about books, and one extra picture is four extra books. This is the most useful conversation on the whole sheet — worth having even if they got it right.

4. How many bikes were in the rack on Wednesday?

Wednesday's bar is the tallest. Its top sits exactly on a grid line. The scale is labelled 0, 2, 4, 6, 8, 10, 12, 14, so the line the bar reaches is the one labelled 12.

Bikes on Wednesday: $\boxed{12}$.

If they say 6: they have counted grid lines (1, 2, 3, 4, 5, 6) instead of reading the number printed beside the line. Both are “counting”, which is why it is such a natural mistake. Ask them to read the number written at the top of the bar's line out loud.

5. How many bikes were in the rack in the whole week?

Read all five bars first and write the numbers down — the addition is the easy part.

Mon $\boxed{8}$, Tue $\boxed{7}$, Wed $\boxed{12}$, Thu $\boxed{10}$, Fri $\boxed{4}$.

Adding: $8 + 7 = 15$, $15 + 12 = 27$, $27 + 10 = 37$, $37 + 4 = 41$.

Bikes altogether: $\boxed{41}$.

A friendlier order. Pairs that make tens first: $8 + 12 = 20$, then $10 + 7 + 4 = 21$, and $20 + 21 = 41$. If the child finds a shortcut like that, it is worth more than the answer.

6. Tuesday is tricky.

Tuesday's bar stops between the grid line labelled $\boxed{6}$ and the grid line labelled $\boxed{8}$ — exactly halfway.

Each gap on this scale is worth 2 bikes, so half a gap is worth 1 bike. Halfway between 6 and 8 is $6 + 1$.

Bikes on Tuesday: $\boxed{7}$.

If they say 8 (or 6): they have rounded the bar up to the nearest labelled line rather than reading where it actually stops. Lay a ruler across the top of the bar to the scale and let them see that it lands between two numbers. On a scale of twos, “between” always means the odd number in the middle.

7. Dev's half picture.

A whole picture is worth $\boxed{4}$ books, so a half picture is worth half of four, which is $\boxed{2}$ books.

Dev's row: two whole pictures and one half. $4 + 4 + 2 = \boxed{10}$.

If they say 12: they have counted the half picture as a whole one — three pictures, three fours. It is the commonest slip on scaled picture graphs, and it is worth asking “is that last one the same size as the others?” rather than saying it is wrong.

If they say 3 or 2.5: they are still counting pictures rather than books. Go back to the key.

8. Talk it out. (Open explanation — listen to the reason, not the number.)

Jo counted the pictures and stopped there. What Jo forgot is the **key** at the bottom of the graph: one whole book picture stands for 4 books, not 1 book. So five pictures means five groups of four, and $5 \times 4 = 20$. Ana read 20 books.

Answers that earn full credit name the key, or describe the counting: “you have to go 4, 8, 12, 16, 20”, “each picture is 4 books so you times it by 4”, “the little writing at the bottom says one picture is 4 books”.

Answers to push on gently: “Jo counted wrong” (Jo counted the pictures perfectly — the pictures are not what the question asked about), and “Jo should have looked at the graph better” (which part of it?). Ask “what would Ana’s row look like if one picture meant one book?” — twenty pictures — and let them see why the key exists.

A good sign: a child who says the graph would be too long to fit without a key has understood what scaling is *for*, which is more than the sheet asked.

Open explanation — open response.